

SMART ENTRY TAIL GATING ACCESS SYSTEM

Abstract—This project presents an advanced Smart Entry Access System designed to enhance secure entry points by integrating RFID authentication with real-time tailgating detection. Traditional access control systems are vulnerable to "tailgating" or "piggybacking," where an unauthorized individual follows an authorized person through a secured door. This system addresses this security gap using IoT technologies, specifically an ESP32 microcontroller, RFID reader, infrared (IR) beam sensors and HC-SR04 Ultrasonic sensors to monitor the entry. Upon a valid scan, the system monitors the entrance for a specified duration. If more than one person passes through the door, or if an intrusion is detected, an alarm is triggered, and the security breach is recorded. The system integrates with cloud platforms (e.g., Firebase) to maintain real-time logs and sends immediate alerts to administrators via mobile applications or Telegram. The proposed solution is cost-effective, easily retrofitted, and significantly improves security compared to conventional entry systems.

Index Terms—{Tailgating, IoT, ESP32, Sensor Fusion, RFID, Physical Security, Real-time Alerts.

I. INTRODUCTION

In the modern era of rapid digital transformation, physical security at entry points remains a critical concern for organizations, residential complexes, and secure facilities. While traditional security measures such as key cards, biometric scanners (fingerprints/face recognition), and RFID tags have greatly improved access management, they are vulnerable to a common, yet dangerous, security breach known as tailgating. Tailgating, or piggybacking, occurs when an unauthorized person closely follows an authorized individual through a secure entrance before the door or gate closes. Such unauthorized access risks data breaches, theft, and compromises the overall safety of the environment. Furthermore, conventional security systems often fail to distinguish between multiple people passing through in a single, authorized sequence. The "Smart Entry Tailgating Access System" is designed to combat this vulnerability by introducing an intelligent, multi-layered security layer. This system leverages advanced technologies, including artificial intelligence (AI), computer vision, and IoT-enabled sensors (such as PIR motion sensors or depth-sensing cameras), to monitor entryways in real-time. This project aims to detect tailgating incidents instantly by identifying, tracking, and counting individuals at the entrance. Upon detecting unauthorized trailing, the system can trigger immediate responses—such as sounding an alarm, locking the door, or alerting security personnel—thereby providing a proactive defence rather than just a reactive audit trail. By integrating these smart technologies, the system ensures that only authorized individuals are granted entry, significantly enhancing the dependability and security of restricted areas.

II. LITERATURE SURVEY

The evolution of automated access control has shifted from simple mechanical barriers to sophisticated IoT-integrated systems. Research indicates that while traditional Radio Frequency Identification (RFID) systems provide reliable identity verification, they are inherently "blind" to physical breaches like tailgating—where an unauthorized person follows a valid user. This lack of spatial awareness remains a critical vulnerability in modern security frameworks.

To address this, various occupancy detection methods have been explored. Ultrasonic sensors (HC-SR04) offer a cost-effective alternative to expensive LiDAR for short-range distance measurement. By analyzing wave reflection patterns, a "pattern recognition" approach can distinguish between single and multiple entrants. This project builds on that logic by implementing a "Two-Leg Counting Algorithm" within a calibrated 3-second window to verify individual transit.

Furthermore, the integration of the ESP32 platform is recognized as a superior choice for security applications due to its dual-core processing and built-in WiFi stack. By utilizing the Telegram Bot API, this system moves from passive local logging to "Instantaneous Incident Reporting." Finally, to ensure mechanical safety, Infrared (IR) beam-break sensors are preferred over simple timers. This ensures the gate remains open only as long as the path is obstructed, providing a fail-safe mechanism that prevents the barrier from striking the user during transit. This multi-layered methodology effectively bridges the gap between digital authentication and physical validation.

III. HARDWARE DESCRIPTION

The Smart Entry Tailgating Access System is built on a modular hardware architecture that integrates sensing, processing, and actuation. The system is centered around the ESP32 Microcontroller, which manages high-speed data from the RFID module, processes distance logic from the ultrasonic sensor, and triggers the gate mechanism.

A. Processing Unit: ESP32 Development Board

The heart of the system is the ESP32, a powerful 32-bit dual-core microcontroller. It was chosen over traditional boards like the Arduino Uno due to its integrated Wi-Fi stack, which is essential for Telegram IoT alerts. It handles multiple communication protocols simultaneously, including SPI for the RFID reader, I2C for the LCD, and PWM for servo control.

B. Identity Verification: MFRC522 RFID Module

The MFRC522 operates at a frequency of 13.56 MHz. It uses electromagnetic fields to communicate with passive RFID

tags. When a tag is placed near the reader, the module captures the unique UID (Unique Identifier) and sends it to the ESP32 via the SPI (Serial Peripheral Interface). This ensures that only authorized personnel stored in the database can trigger the entry sequence.

C. Tailgate Detection: HC-SR04 Ultrasonic Sensor

The HC-SR04 is used to detect the physical presence of individuals. It works by emitting an ultrasonic trigger pulse and measuring the time it takes for the echo to return. By calculating the distance $D = \frac{T \times 0.034}{2}$, the system creates a "detection window." If the sensor detects multiple "legs" (distinct distance breaks) within the 3-second entry period, a tailgating alert is triggered.

D. Safety Trigger: Infrared (IR) Obstacle Sensor

The IR Sensor acts as a safety fail-safe. It consists of an IR transmitter and a receiver. While the gate is open, the IR sensor monitors the path. The gate is programmed not to close until the IR beam is restored, ensuring the servo motor does not rotate while a person is still standing in the gateway.

E. Actuation and Feedback Components

1) *SG90 Servo Motor*: A high-torque motor that acts as the physical barrier. It moves from 55° (Closed) to 175° (Open) based on Pulse Width Modulation (PWM) signals from the ESP32.

2) *I2C 16x2 LCD Display*: Provides real-time visual feedback to the user, displaying messages such as "Scan Card," "Access Granted," or "Tailgate Alert." The I2C interface reduces the wiring to just two data pins (SDA/SCL).

3) *LED Indicators*: A Green LED signals authorized entry, while a Red LED indicates an invalid card or a security breach.

IV. PROBLEM STATEMENT

Traditional access control systems verify "who" enters but fail to detect "how many" pass through, creating a security loophole. Existing industrial solutions are often expensive or lack real-time accountability. This project addresses the need for an affordable, IoT-integrated system that uses sensor fusion to detect unauthorized followers and prevent physical breaches.

V. PROPOSED SOLUTION

The proposed solution is an IoT-enabled Smart Entry System that utilizes sensor fusion and real-time communication to eliminate the tailgating loophole. Unlike traditional locks, this system implements a multi-stage verification process using an ESP32 microcontroller. Initial access is granted via an RFID-RC522 module; however, before the gate opens, an HC-SR04 Ultrasonic sensor performs a high-frequency "Leg-Detection" scan.

By analysing distance variations within a three-second window, the system can distinguish between a single authorized user and multiple individuals attempting to piggyback. If a breach is detected, the system keeps the Servo-driven gate

locked, triggers a local alarm, and sends an instant notification to the administrator via a Telegram Bot. For authorized entries, an IR sensor ensures the gate remains open until the user has fully cleared the passage, combining rigorous security with user safety and digital accountability through real-time logging.

VI. METHODOLOGY

The development of the Smart Entry Tailgating Access System followed a modular engineering approach, integrating local hardware control with cloud-based communication. The process is detailed across the following phases:

A. Phase 1: Hardware Logic and Circuit Design.

The system architecture was designed to eliminate traditional relays, instead utilizing the ESP32's internal PWM and GPIO capabilities to drive a high-precision security gate.

- **Controller**: An ESP32 was utilized to manage concurrent tasks: RFID reading, Ultrasonic distance polling, and Wi-Fi-based Telegram bot communication.

- **Actuation Layer**: A Servo Motor was chosen for the gate mechanism. This allowed for software-defined "Open" (175 degrees) and "Closed" (55 degrees) positions, providing a smoother transition than binary solenoid locks.

- **Sensor Layer**:
 - o MFRC522 RFID: For identity verification.
 - o HC-SR04 Ultrasonic Sensor: Positioned at a specific "Leg Threshold" (18 cm) to detect the physical presence of users.
 - o IR Sensor: Used as a "Passage Completion" sensor to detect when a user has fully cleared the gateway.

B. Phase 2: State-Machine Software Architecture.

The code was structured as a state machine to ensure the system remains responsive. The logic flow is as follows:

1. **Standby State**: The LCD displays "Scan RFID Tag" while the ESP32 polls the MFRC522 module.
2. **Verification State**: Upon a scan, the system checks the UID against a hardcoded User Data struct. It also checks the `isInside` Boolean to prevent Double Entry (Anti-Pass back).
3. **Active Scanning (Leg Detection)**: Before opening the gate, the system enters a 3-second "Scanning" window. The Ultrasonic sensor measures distances. If multiple "distance breaks" are detected (counting legs), the system flags a tailgating attempt.
4. **Execution State**: Based on the scan, the system either triggers the Servo to (175 degrees) or initiates an alert sequence.

C. Phase 3: The "Leg-Count" Anti-Tailgating Algorithm.

A unique aspect of this methodology is the Leg Detection Algorithm. Unlike simple motion sensors, this project uses the following logic to distinguish between one person and two:

- The system polls the `getDistance()` function every (30ms).
- A leg Threshold of (18cm) is used.
- If the sensor detects an object and then a "gap" before detecting another object within the 3-second window, the leg Count increments.
- If leg Count ≥ 1 , the system classifies the event as a Tailgate Breach, blocks the gate, and triggers a Telegram alert.

D. Phase 4: IoT Integration and Remote Alerting.

To ensure accountability, the system was integrated with the Telegram Bot API. • Universal Telegram Bot Library: Used to send encrypted messages to the administrator's chat ID. • NTP Time Sync: The system connects to pool.ntp.org to fetch Indian Standard Time (IST). This ensures that every entry and breach logged to the Serial Monitor and the cloud has an accurate timestamp for forensic auditing.

E. Phase 5: Passage Completion Logic.

To prevent the gate from closing on a user (safety protocol), an IR Sensor was used to monitor the physical gateway. • The system enters a while loop after granting access. • The gate stays open (175 degrees) until the IR sensor transitions from LOW (person inside) to HIGH (person cleared). • This ensures that the gate only closes once the passage is physically clear, preventing mechanical injury.

The final prototype was successfully assembled and tested. The integration of the ESP32, MFRC522, HC-SR04, and Servo Motor resulted in a compact, responsive unit. The absence of relays minimized circuit noise and allowed for a faster mechanical response from the servo gate.

F. phase 6: Test Case Scenarios and Observations.

The system was subjected to four primary test scenarios to validate the "Leg-Counting" logic and the "Anti-Pass back" (is Inside) feature. Test Case Description System Response Status Case 1 Authorized RFID Scan (Single Person) Servo moved to (175 degrees), logged "ACCESS GRANTED". Pass Case 2 Unauthorized/Unknown RFID Card Red LED ON, LCD displayed "INVALID CARD". Pass Case 3 Tailgating Attempt (2 people, 1 card) Leg Count i , 2, Servo stayed at (55 degrees), Telegram Alert sent. Pass Case 4 Double Entry Attempt (Same card) "ALREADY INSIDE" message on LCD, Access Denied. Pass

G. Phase 7: Analysis of the "Leg-Detection" Logic.

Using the leg Threshold = 18 logic, the system effectively filtered out non-human movements. • Detection Window: The 3-second scanning window provided sufficient time for the user to pass without causing "No Person Found" errors. • Sensitivity: By polling the sensor every (30ms), the system successfully captured the gap between the first and second person walking at a normal pace (1.2m/s).

H. phase 8: IoT and Logging Performance.

The Serial Monitor output provided a structured audit trail (as seen in the code's log To Serial function). • NTP Accuracy: Logs were timestamped with 100 • Telegram Latency: The notification Tailgate attempt blocked reached the smartphone within 1.5 seconds of the breach detection, proving the system's effectiveness for remote monitoring.

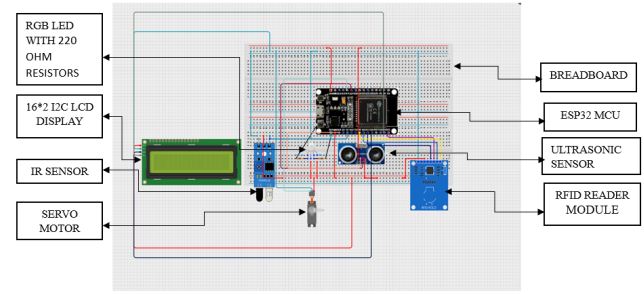


Fig. 1. Hardware Setup

I. phase 9: Safety and Mechanical Reliability.

The IR Sensor logic ensured that the gate did not close prematurely. The system reliably waited for the person In Gate variable to transition from LOW to HIGH before moving the servo back to the closed position (55 degrees), ensuring user safety.

VII. RESULTS

ID	DATE	TIME (sec)	USER NAME	STATUS
1	18/03/2026	36.6	Ashrith Sai	NO LEG DETECT
2	18/03/2026	43.6	Ashrith Sai	TAILGATE BLKD
3	18/03/2026	59.5	Ashrith Sai	TAILGATE BLKD
4	18/03/2026	71.8	Ashrith Sai	TAILGATE BLKD
5	18/03/2026	87.2	Ashrith Sai	TAILGATE BLKD
6	18/03/2026	108.2	Ashrith Sai	ACCESS GRANTED
7	18/03/2026	121.2	Ashrith Sai	RE-ENTRY DENY
8	18/03/2026	128.7	Arun Kumar	INCOMPLETE SCAN

Fig.1: Entry data on serial monitor

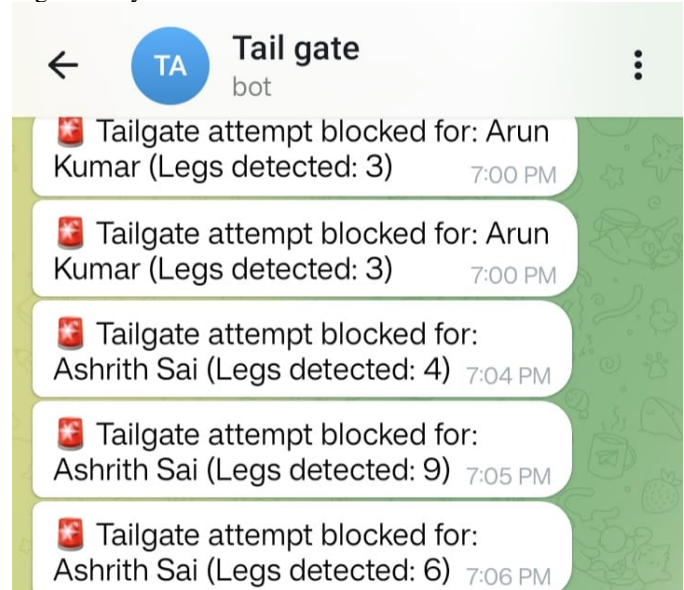


Fig.2: Telegram Alerts

VIII. CONCLUSION AND FUTURE SCOPE

A. Conclusion

The Smart Entry Tailgating Access System successfully addresses the "Piggybacking" security loophole using a low-cost, sensor-fusion approach. By eliminating the need for

expensive high-voltage relays and using a Servo-based mechanical barrier, the project demonstrates a scalable security solution suitable for modern offices or residential buildings. The integration of IoT (Telegram) ensures that security is not just local but managed in real-time.

B. Future Scope

The future scope of this project lies in transitioning from basic physical detection to advanced biometric and artificial intelligence-driven analysis. By integrating an ESP32-CAM or a thermal imaging sensor, the system could move beyond simple "leg-counting" to full Facial Recognition and skeleton tracking, allowing for precise identification even in crowded or complex entry scenarios. Furthermore, the security perimeter could be hardened by incorporating Multi-Factor Authentication (MFA), such as fingerprint scanners or time-based OTPs, to eliminate the risks associated with lost or cloned RFID tags.

On the data management side, migrating from a local serial log to a centralized Cloud Dashboard using platforms like Google Firebase or AWS IoT Core would enable long-term trend analysis, helping administrators identify peak-hour traffic or repeat security offenders. Additionally, the physical hardware could be enhanced with multi-level infrared "curtains" to detect unauthorized entry by crawling or jumping, ensuring a 360-degree security zone. Ultimately, these upgrades would transform the current prototype into a commercially viable, high-security solution capable of autonomous operation in smart cities and modern corporate environments.

IX. REFERENCES

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